A Near-Logarithmic Approximation for All-or-Nothing Multicommodity Flow

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Abstract

In the all-or-nothing multicommodity flow problem we are given an undirected graph with unit edge capacities and k terminal pairs, and we seek a largest set of pairs that can simultaneously send one unit of (splittable) flow each. The best known approximation ratio, due to Chekuri, Khanna and Shepherd (STOC 2005), is $O(\log^2 k)$, and Bansal, Jambulapati and Saranurak (ICALP 2026) conjectured that a $\log^{1+o(1)} k$ approximation exists. We confirm this conjecture: we give a polynomial-time algorithm that routes at least $\mathrm{LP}/(\log k \cdot 2^{O(\sqrt{\log\log k})})$ pairs with congestion one, where LP is the value of the natural multicommodity flow relaxation. Hence the integrality gap of this relaxation is $\log^{1+o(1)} k$; the known lower bound is $\log^{1/2-\delta} n$.

The proof has two parts. First, a single decomposition computed by the integral-node-weight flow-expander decomposition algorithm of Bansal, Jambulapati and Saranurak, with node weights obtained by discretizing the fractional solution at granularity $\mathrm{LP}/(2k)$, yields the well-linked decomposition of Chekuri, Khanna and Shepherd with loss $\log^{1+o(1)} k$ instead of $O(\log^2 k)$. This is the decomposition asked for by Bansal, Jambulapati and Saranurak, and the argument is short. Second, we give a complete and self-contained proof of the congestion-one rounding inside flow-linked instances that the decomposition requires. This rounding was stated by Chekuri, Khanna and Shepherd in 2005 together with a remark that its clustering had to be strengthened; the strengthened argument appeared in their 2013 journal paper, phrased for Räcke-tree instances. Our proof follows their architecture, replaces cluster centers by a min-cut criterion, and uses a degree reduction to make clusters vertex-disjoint. We also extend the rounding to non-uniform paired weights.

1 Introduction

The problem. Let $G = (V, E)$ be an undirected graph with unit edge capacities and let $(s_1, t_1), \ldots, (s_k, t_k)$ be (indexed) terminal pairs. A set $I \subseteq [k]$ is *routable* if there is a multicommodity flow that sends one unit from s_i to t_i for every $i \in I$ and whose total load on every edge is at most one; flow of a commodity may be split along many paths. The *all-or-nothing multicommodity flow problem* (AN-MCF) asks for a routable set of maximum cardinality. It was introduced by Chekuri, Mydlarz and Shepherd [8] and studied in general graphs by Chekuri, Khanna and Shepherd [5, 7], as a relaxation of the edge-disjoint paths problem (EDP) that, unlike EDP, admits polylogarithmic approximation ratios: the natural multicommodity flow relaxation has an $\Omega(\sqrt{n})$ integrality gap for EDP but only a polylogarithmic one for AN-MCF. Call $x \in [0,1]^k$ *fractionally routable* if some multicommodity flow with load at most one on every edge routes x_i units between s_i and t_i for every i. The relaxation is

$$\mathrm{LP} = \max\Big\{\textstyle\sum_{i=1}^{k} x_i \;:\; x \in [0,1]^k \text{ fractionally routable}\Big\}; \tag{1}$$

it has a polynomial-size edge formulation, and clearly $\mathrm{LP} \geq \mathrm{OPT}$.

Prior work. Chekuri, Khanna and Shepherd [5, 7] gave the first polylogarithmic approximation, $O(\log^3 n \log\log n)$, using Räcke's hierarchical decompositions [14, 11]. In [6] they made precise the notion of a *well-linked decomposition* (used earlier in [5] and in work on planar EDP) and introduced its weighted version: a fractional solution is converted into a collection of vertex-disjoint instances in each of which the terminals are fractionally flow-linked, losing a factor $O(\beta \log k)$, where $\beta = O(\log k)$ is the flow-cut gap for product multicommodity flows [12, 13, 3]; inside a flow-linked instance, a congestion-one rounding loses only a constant [6, Theorem 3.1]. This gives the current best ratio $O(\log^2 k)$ for general graphs [6, Theorem 3.2] (and $O(\log k)$ in planar graphs). On the negative side, AN-MCF is hard to approximate within $\log^{1/2-\delta} n$ for every $\delta > 0$ unless NP $\subseteq$ ZPTIME($n^{\mathrm{polylog}\, n}$), and the relaxation (1) has integrality gap $\log^{1/2-\delta} n$ [2]; the problem is APX-hard already on trees [10].

Recently Bansal, Jambulapati and Saranurak [4] gave the first polynomial-time flow-expander decompositions with overhead $\log^{1+o(1)}$, circumventing the flow-cut gap. In their list of open questions [4, Section 5] they observe that the $O(\log^2 k)$ bound for AN-MCF is the product of the flow-cut gap and the recursion depth of the well-linked decomposition, and that their technique suggests these might be telescoped; but they point out that the connection is not black-box, since the well-linked decomposition of [6] interleaves concurrent-flow computations with maximum-throughput flow steps. They ask whether the decomposition of [6] can be built with $\log^{1+o(1)} k$ overhead, note that this would immediately imply the following conjecture, and state:

> **Conjecture 5.1 of [4].** There is a polynomial-time $\log^{1+o(1)} k$-approximation algorithm for the all-or-nothing flow problem.

1.1 Our results

All logarithms are binary. We fix, once and for all, a nondecreasing function $\Gamma : \mathbb{Z}_{\geq 0} \to \mathbb{R}_{\geq 1}$ with

$$\Gamma(a) = O\big(\log(a+2) \cdot \exp\big(O(\sqrt{\log\log(a+4)})\big)\big) \tag{2}$$

such that the decomposition theorem of [4] holds with cut bound $\phi|A|\Gamma(|A|)$ (Theorem 2.3 and the discussion after it).

Theorem 1.1 (Main theorem)**.** *There is a polynomial-time algorithm that, given an undirected graph with unit edge capacities and k terminal pairs, outputs a routable set of pairs together with a routing of congestion one, of size at least*

$$\frac{\mathrm{LP}}{c\,\Gamma(4k)} \;\geq\; \frac{\mathrm{OPT}}{C \log(k+2) \exp\big(C\sqrt{\log\log(k+4)}\big)},$$

where c, C are absolute constants. In particular the integrality gap of (1) *is* $\log^{1+o(1)} k$.

This confirms Conjecture 5.1 of [4]. The guarantee depends only on k (not on the number of vertices), no congestion is violated, and shared endpoints, coincident endpoints and repeated pairs are allowed. Apart from the call to the decomposition algorithm of [4], the algorithm is deterministic. Theorem 1.1 is the composition of the following two results.

Well-linked decomposition with $\log^{1+o(1)} k$ loss. Following [6], for a set X of vertices and weights $\pi : X \to \mathbb{R}_{\geq 0}$, X is *π-flow-linked* in G if the product demand $\pi(u)\pi(v)/\pi(X)$ between every two distinct $u, v \in X$ is routable with congestion one.

Theorem 1.2 (see Theorem 4.1)**.** *Given a feasible fractional multicommodity flow of total value F for k pairs with distinct degree-one endpoints, one can compute in polynomial time vertex-disjoint induced subgraphs $G_1, \ldots, G_\ell$ and, for each j, a set P_j of pairs inside G_j with weights π_j on their endpoints such that $\pi_j(s_i) = \pi_j(t_i) \le 1$, the endpoints X_j of P_j are π_j-flow-linked in G_j, and $\sum_j \pi_j(X_j) \ge F/(c\,\Gamma(4k))$.*

This is the well-linked decomposition of [6, Theorem 2.1] with the loss $O(\log^2 k)$ replaced by $\log^{1+o(1)} k$, i.e. the decomposition asked for in [4, Section 5]. As anticipated in [4], Theorem 1.1 follows by combining it with congestion-one rounding inside flow-linked instances; we now discuss this second step, which is where most of the work of this paper lies.

Congestion-one rounding, with a complete proof. The rounding step is the following statement [6, Theorem 3.1]: if X is α-flow-linked (i.e. $\pi \equiv \alpha$), then for every matching M on X one can route $\Omega(\alpha|M|)$ pairs of M with congestion one. In [6] this theorem is attributed to the conference version [5] of [7], with a footnote stating that the clustering of [5] needs to be strengthened for congestion one and that details would appear in the journal version. The journal version [7, Section 3.4] contains the strengthened clustering. Its theorem [7, Theorem 3.8] is stated for "nice" instances defined through a Räcke tree, which is used to obtain a flow to a common distribution; the general case of clusters sharing vertices is indicated briefly at the end of the section, and the weighted case is outlined. Since the rounding is the step where congestion exactly one is obtained, we give a complete self-contained proof for arbitrary flow-linked sets (Section 3).

Theorem 1.3 (Theorem 3.1 and Corollary 3.2)**.** *(a) If X is α-flow-linked in G, $0 < \alpha \le 1$, and M is a matching on X, then one can compute in polynomial time a set of at least $\max\{1, \alpha|M|/2^{15}\}$ pairs of M that is routable with congestion one.*

(b) If the vertices of X have degree one, M is a perfect matching on X, $\pi : X \to [0,1]$ satisfies $\pi(s) = \pi(t)$ for $st \in M$, and X is π-flow-linked, then one can compute in polynomial time a set of at least $\max\{1, \pi(X)/2^{19}\}$ pairs of M routable with congestion one.

Part (b), a "non-uniform" rounding, follows from part (a) by the copying device of [6, Lemma 1.1], in which the unit edges at the original terminals act as admission bottlenecks.

Other consequences and limits. The reduction behind Theorem 1.2 accepts any flow-expander decomposition algorithm for integral node weightings. Anand [1] recently improved flow-expander decomposition to overhead $O(\log n \log\log n)$. His bound in terms of the total weight $|A|$ requires every vertex to have weight at least one, which forces $|A| \ge n$, so we were not able to obtain an $O(\log k \log\log k)$ ratio. If the node-weighted bound stated in Appendix C of [1] (zero weights allowed, overhead $O(\log n \log\log n)$) is used, the same reduction gives an $O(\log n \log\log n)$-approximation (Remark 5.4); this is conditional on that statement and is not a bound in terms of k. For the weighted (profit) version we only obtain $\log^{2+o(1)} k$ (Remark 5.3); a $\log^{1+o(1)} k$ bound for it, and the node-capacitated version, remain open (Section 6).

1.2 Technical overview

The decomposition. The decomposition of [6] fixes a path decomposition of the fractional solution and recursively cuts sparse cuts with respect to the current weights $\gamma(u, H)$, the flow of paths fully contained in the current piece H. These weights shrink whenever a cut destroys paths, so the weighting changes during the recursion, whereas the decomposition of [4] works with one fixed integral node weighting A and charges the cut to $|A|$. We therefore fix the weights in advance.

We discretize the fractional solution x at granularity $\varepsilon = F/(2k)$: both endpoints of pair i receive the integral weight $d_i = \lfloor x_i/\varepsilon \rfloor$ (zero if $x_i < \varepsilon$), in the spirit of the uniformization of [6, Lemma 1.1]. Then $K = \sum_i d_i \le 2k$, so $|A| \le 4k$, while $\varepsilon K \ge F/4$. We apply [4] once (up to a constant number of trials to calibrate an unknown constant), with $\phi = \Theta(\varepsilon/\Gamma(4k))$, so that at most $\varepsilon K/4$ edges are cut. Two simple observations finish the proof. First, a pair whose endpoints are separated sends all of its flow $x_i \ge \varepsilon d_i$ across the cut, so separated pairs carry at most a quarter of the total weight; this is the charging of [6, Lemma 2.2], now against a fixed weighting. Second, flow expansion with respect to A implies flow-linkedness of every sub-collection of weighted terminals, so after discarding separated pairs the remaining endpoints in each component are ϕA-flow-linked, and no weight is ever recomputed.

Admission. The pieces carry non-uniform weights ϕd_i. To use the uniform rounding we rediscretize and attach pendant copies of each terminal; since the original terminals have degree one, two copies of the same pair would have to share the unit edge at s_i, so a congestion-one routing of copy pairs uses at most one copy of each pair, and the rounding does not need to know about copies. This gives Theorem 1.3(b).

The rounding. Our proof of Theorem 1.3(a) follows the architecture of [7, Section 3.4]: terminals are grouped into clusters of $\Theta(1/\alpha)$ terminals, a sparse set of pairs is selected, every selected terminal spreads one unit to the terminals of its (possibly enlarged) cluster using at most half of the capacity of every edge except bridges, the received flow is sent to a common distribution using the other half of the capacity, and the excess on bridges is removed by an integrality argument [7, Lemma 3.10]. We make three changes that allow a short and complete proof: (i) a max-flow/min-cut argument (Lemma 3.8) gives a simple sufficient condition for a terminal to spread its unit, which removes the need for cluster centers; (ii) a flow-preserving degree reduction makes all clusters vertex-disjoint, which removes the case analysis caused by shared connector vertices; (iii) the common-distribution flow is obtained directly from flow-linkedness rather than from a Räcke tree.

On the contribution. Once Theorem 3.1 of [4] is available, the decomposition (Theorem 1.2) is a short argument, and [4] anticipated that such a decomposition would imply their conjecture. The longer part of the paper is the congestion-one rounding; its structure follows [7, Section 3.4], and our contribution there is a complete proof for arbitrary flow-linked sets, with the simplifications above, and the non-uniform extension. We view the paper as settling the conjecture of [4] by a short reduction together with a careful, self-contained treatment of the rounding step that the reduction needs.

Organization. Section 2 gives definitions and the tools we use. Section 3 proves the rounding theorem, Section 4 the decomposition theorem, and Section 5 the main theorem and its corollaries. Section 6 discusses lower bounds and open problems.

2 Preliminaries

Graphs and flows. Graphs are undirected and may have parallel edges; every edge has capacity one. For a graph H and $Z \subseteq V(H)$, $\delta_H(Z)$ is the multiset of edges with exactly one endpoint in Z, and $H[Z]$ is the induced subgraph. An edge e of H is a *bridge of* H if $H - e$ has more connected components than H; an edge with a parallel copy is never a bridge. If e is a bridge of a connected graph H, then $H - e$ has exactly two components. A flow from u to v of value ν is a nonnegative

function on the arcs of the bidirected version of H satisfying conservation at all vertices other than u, v, with net outflow ν at u. The load of a collection of flows on an edge e is the sum of all their values on the two arcs of e. A demand D (a nonnegative function on unordered pairs of distinct vertices) is *routable with congestion* κ if there are flows of value $D(u,v)$ between u and v for all pairs with total load at most κ on every edge. A set of terminal pairs is *routable* if the demand placing one unit on each pair is routable with congestion one. Routability of a given set is decidable by linear programming. If Z contains u but not v, then every flow from u to v of value ν has load at least ν on $\delta_H(Z)$.

Definition 2.1 (flow-linked sets [6])**.** Let $X \subseteq V(H)$ and $\pi : X \to \mathbb{R}_{\geq 0}$ with $\pi(X) > 0$. Then X is *π-flow-linked in H* if the demand $\pi(u)\pi(v)/\pi(X)$ on every pair of distinct $u, v \in X$ is routable in H with congestion one. For a constant α, X is *α-flow-linked* if it is π-flow-linked for $\pi \equiv \alpha$, i.e. the demand $\alpha/|X|$ on every pair is routable with congestion one.

If X is π-flow-linked in H and $|X \cap \mathrm{supp}(\pi)| \geq 2$, then all vertices of X with positive weight lie in one component of H.

Definition 2.2 (flow expansion [4])**.** Let $A : V(H) \to \mathbb{R}_{\geq 0}$. A demand D *respects* A if $\sum_{y \neq x} D(x,y) \leq A(x)$ for every vertex x. The weighting A is *ϕ-flow-expanding in H* if every demand that respects A is routable in H with congestion $1/\phi$. For $U \subseteq V(H)$, $A|_U$ denotes the restriction of A to U.

Theorem 2.3 (Bansal, Jambulapati and Saranurak [4, Theorem 3.1])**.** *For every undirected graph H, every integral node weighting $A : V(H) \to \mathbb{Z}_{\geq 0}$ and every $\phi > 0$, one can compute in polynomial time a set $C \subseteq E(H)$ such that $|C| \leq \phi |A| \log(|A|) \exp(O(\sqrt{\log\log |A|}))$ and, for each connected component U of $H - C$, the weighting $A|_U$ is ϕ-flow-expanding in $H[U]$.*

We use this theorem as a black box, with the following conventions.

- *The overhead function.* We write the bound as $|C| \leq \phi|A|\Gamma(|A|)$ with $\Gamma = c_0\Gamma_1$, where $\Gamma_1(a) = \lceil \log(a+4) \rceil \cdot 2^{\lceil \sqrt{\log\log(a+16)} \rceil}$ is explicit and $c_0 \geq 1$ is the absolute constant implicit in [4]. Our algorithm does not need to know c_0 (see the proof of Theorem 4.1). For $|A| \leq 1$ every respecting demand is zero (a respecting demand on $\{x, y\}$ is at most $\min\{A(x), A(y)\}$), so $C = \emptyset$ suffices. The internal parameters of the algorithm in [4] are meant for large $|A|$; for $|A|$ bounded by an absolute constant one may instead use classical cut-and-recurse, which with the $O(\log |A|)$ flow-cut gap for product demands [12, 13, 3] has overhead $O(\log^2 |A|) = O(1)$. We take c_0 large enough to cover these cases.
- *A factor of two.* The stopping rule of the algorithm in [4] combined with their Fact 2.1 literally certifies $(\phi/2)$-flow expansion; if only this is available, one calls the algorithm with 2ϕ, which changes c_0 by a factor two. We use the theorem as stated.
- *Multigraphs.* Parallel edges can be subdivided without changing any flow or cut quantity.
- *Randomness.* The algorithm of [4] is built from linear programming, sweep cuts and the sparse neighborhood cover of their Lemma 3.2; we have not verified whether the latter is deterministic, and we use no property of the algorithm beyond the statement above. All other steps of our algorithms are deterministic.

The following fact connects flow expansion to flow-linkedness.

Lemma 2.4. *Let A be ϕ-flow-expanding in H and let $Y \subseteq V(H)$ with $A(Y) > 0$. Then Y is π-flow-linked in H for $\pi = \phi A|_Y$.*

Proof. The π-product demand $D(u,v) = \phi A(u)A(v)/A(Y)$ $(u \neq v \in Y)$ has marginal at most $\phi A(u)(A(Y) - A(u))/A(Y) \leq \phi A(u)$ at $u \in Y$ and is zero elsewhere. So D/ϕ respects A and is routable with congestion $1/\phi$; scaling by ϕ routes D with congestion one. □

The next lemma is [7, Lemma 3.10]; we include its short proof.

Lemma 2.5 (bridge integrality)**.** *Let H be a graph and N a set of terminal pairs. Suppose flows of value one for all pairs in N have total load at most one on every non-bridge of H and strictly less than two on every bridge of H. Then N is routable in H with congestion one, and such a routing can be computed in polynomial time.*

Proof. For each pair, decompose its flow into flows along paths and cycles and discard the cycles; loads do not increase, and each pair is now routed along simple paths. If e is a bridge, a simple path from s to t uses e exactly once if e separates s from t and not at all otherwise. Hence the load on e is the number of pairs of N separated by e, an integer smaller than two, i.e. at most one. □

Normalized instances. An AN-MCF instance is *normalized* if its $2k$ terminals are distinct vertices of degree one. Every instance can be normalized: for every pair i add two new vertices s_i', t_i' and unit edges $s_i s_i'$, $t_i t_i'$, and replace the pair by (s_i', t_i'). Unit flows extend along the new edges (each is used only by its own pair) and project back by deleting them, so routable sets, OPT and LP are unchanged (a pair with $s_i = t_i$ becomes a pair at distance two). Parallel edges of the input may be subdivided. Hence we assume throughout that the input is normalized and simple.

3 Congestion-one rounding in flow-linked instances

Theorem 3.1. *Let G be a connected graph, $X \subseteq V(G)$ a set of $q \geq 2$ vertices, M a matching on X (a non-empty set of pairwise disjoint pairs of elements of X), and $0 < \alpha \leq 1$, such that X is α-flow-linked in G. Then one can compute in polynomial time a set $M' \subseteq M$ with $|M'| \geq \max\{1, \alpha|M|/2^{15}\}$ and a routing of M' with congestion one.*

Corollary 3.2. *Let G be a graph, M a set of pairs whose endpoint set X consists of distinct vertices of degree one, and $\pi : X \to [0,1]$ with $\pi(s) = \pi(t)$ for every $st \in M$ and $\pi(X) > 0$, such that X is π-flow-linked in G. Then one can compute in polynomial time a set $M' \subseteq M$ with $|M'| \geq \max\{1, \pi(X)/2^{19}\}$ and a routing of M' with congestion one.*

As discussed in the introduction, Theorem 3.1 is [6, Theorem 3.1] (with an explicit constant); the proof below follows the strategy of [7, Section 3.4] but is self-contained. Terminals not covered by M are allowed; they serve only as receivers of flow. Sections 3.1–3.5 prove Theorem 3.1, and Section 3.6 proves Corollary 3.2.

3.1 Two reductions

Lemma 3.3. *Given (G, X, M, α) as in Theorem 3.1, one can construct in polynomial time a connected graph $\hat{G}$ and a set $\hat{X} \subseteq V(\hat{G})$ with a bijection $x \mapsto \hat{x}$ from X to $\hat{X}$ such that (a) every vertex of $\hat{X}$ has degree one in $\hat{G}$; (b) every vertex of $\hat{G}$ has at most three distinct neighbours; (c) $\hat{X}$ is α-flow-linked in $\hat{G}$; and (d) if a set of pairs $\{\hat{s}\hat{t}\}$ is routable in $\hat{G}$, the corresponding set $\{st\}$ is routable in G, and the routing can be transformed in polynomial time.*

Proof. First attach to every $x \in X$ a new vertex x' by a unit edge and let $X' = \{x'\}$. The product flow between $u, v \in X$ extends along $u'u$ and vv'; the edge $x'x$ then carries $\alpha(q-1)/q < 1$, so X' is α-flow-linked. Routings project back by deleting the pendant edges.

Next (after deleting loops, which carry no useful flow), let v be a vertex not in X' with incident edges $e_1, \ldots, e_d$, $d \geq 4$ (counted with multiplicity). Replace v by a path $v_1, \ldots, v_d$, attach e_i to v_i, and join each v_l and v_{l+1} by d parallel edges. Do this for every such vertex. Every new vertex has

at most three distinct neighbours, vertices of degree at most three are unchanged, and the graph stays connected. A flow in the new graph projects to the old one by contracting the paths; loads on original edges are unchanged. Conversely, consider flows in the old graph between vertices of X' with load at most one on every edge, decomposed into simple paths. A path through v enters through some e_i and leaves through some e_j; reroute it along $v_i, \dots, v_j$ and split its value evenly among the d parallel edges of each step. The paths crossing the step between v_l and v_{l+1} carry at most $\sum_i \mathrm{load}(e_i) \le d$ in total, so each parallel edge carries at most one. Applying this at every split vertex lifts any routing with congestion one; in particular the product flow, so (c) holds. Vertices of X' have degree one and are not split, so (a) holds. □

By Lemma 3.3 we assume, until the end of Section 3.5, that G is connected, every terminal (vertex of X) has degree one, every vertex has at most three distinct neighbours, and X is α-flow-linked. "Bridge" always means a bridge of this graph G. We set

$$\rho := \alpha/4.$$

3.2 Common distribution and assembly

Lemma 3.4 (common distribution)**.** *There are flows h_v ($v \in X$), where h_v sends ρ/q units from v to every $w \in X \setminus \{v\}$, with total load at most $1/2$ on every edge.*

Proof. Let ψ_{vw} be flows routing α/q between every pair $\{v, w\}$ of distinct terminals with total load at most one. Let h_v consist of $\frac{1}{4}\psi_{vw}$, oriented from v to w, for all $w \neq v$; it sends $\alpha/(4q) = \rho/q$ to w. Each ψ_{vw} is used with factor $\frac{1}{4}$ by h_v and by h_w, so the total load is at most half that of all ψ_{vw}. □

Definition 3.5. For $s \in V(G)$ and $\gamma : X \to \mathbb{R}_{\ge 0}$ with $\sum_v \gamma(v) = 1$, a *flow from s to γ* is a function on arcs whose net outflow at every vertex y equals $[y = s] - \gamma(y)$ (with $\gamma(y) = 0$ for $y \notin X$). Let $S \subseteq X$. A *spreading system* for S consists of flows g_s from s to γ_s, $s \in S$, such that $\sum_{s \in S} \gamma_s(v) \le \rho$ for every $v \in X$, and the total load of all g_s is at most $1/2$ on every non-bridge and at most one on every bridge.

Lemma 3.6 (assembly)**.** *If $M' \subseteq M$ and the set S of endpoints of M' has a spreading system, then M' is routable with congestion one, and a routing can be computed in polynomial time.*

Proof. For $s \in S$ let $\sigma_s := g_s + \sum_{v \in X} (\gamma_s(v)/\rho)\, h_v$. The term $(\gamma_s(v)/\rho)h_v$ moves $\gamma_s(v)/q$ from v to every $w \neq v$ and leaves $\gamma_s(v)/q$ at v, so σ_s is a flow from s to the uniform distribution on X: the net inflow at $w \in X$ is $\gamma_s(w)/q + \sum_{v \neq w} \gamma_s(v)/q = 1/q$. For $st \in M'$, the sum of σ_s and the reversal of σ_t is therefore a flow from s to t of value one. The total load of all these flows is at most the load of $\sum_s g_s$ plus $\sum_{v \in X} \big(\sum_s \gamma_s(v)/\rho\big)\mathrm{load}(h_v) \le \sum_v \mathrm{load}(h_v) \le 1/2$ by Lemma 3.4. Thus the load is at most 1 on non-bridges and at most $3/2 < 2$ on bridges, and Lemma 2.5 gives a routing with congestion one. □

3.3 A criterion for spreading

Definition 3.7. Let Q be a connected subgraph of G, $s \in V(Q)$ and $R \subseteq X \cap V(Q)$. We say that *s spreads in (Q, R)* if there is a flow in Q from s to some γ with $\gamma(v) \le \rho$ for $v \in R$ and $\gamma(v) = 0$ for $v \notin R$, whose load is at most $1/2$ on every edge of Q that is not a bridge of G and at most one on every edge of Q that is a bridge of G.

(Up to normalization, this is the notion of a *pseudocenter* in [7].)

Lemma 3.8 (criterion)**.** *Let Q be a connected subgraph of G, $s \in V(Q)$ and $R \subseteq X \cap V(Q)$ with $\rho|R| \geq 1$. Suppose that for every edge e of Q that is a bridge of Q but not a bridge of G, the component of $Q - e$ containing s contains at least $1/\rho$ vertices of R. Then s spreads in (Q, R), and the flow can be computed by a maximum-flow computation.*

Proof. Give every edge of Q capacity one if it is a bridge of G and $1/2$ otherwise, add a sink z and an arc of capacity ρ from every $v \in R$ to z. It suffices to show that every s–z cut has capacity at least one: then a maximum s–z flow, scaled to value one, is the required flow, with $\gamma(v)$ the flow on the arc vz. Such a cut is given by $Z \subseteq V(Q)$ with $s \in Z$, and its capacity is $c(\delta_Q(Z)) + \rho|R \cap Z|$. If $\delta_Q(Z) = \emptyset$ then $Z = V(Q)$ and the capacity is at least $\rho|R| \geq 1$. If $|\delta_Q(Z)| \geq 2$, then $c(\delta_Q(Z)) \geq 1$. If $\delta_Q(Z) = \{e\}$, then e is a bridge of Q and Z is the component of $Q - e$ containing s; either e is a bridge of G and $c(e) = 1$, or by hypothesis $\rho|R \cap Z| \geq 1$. □

3.4 Clusters, bad bridges and tags

Lemma 3.9 (clusters)**.** *Let $W \geq 1$ with $\rho q \geq W$. One can compute a partition $V(G) = V_1 \cup \cdots \cup V_p$ such that every $G[V_j]$ is connected and $W \leq \rho|X \cap V_j| < 4W$ for every j.*

Proof. Since $\rho q \geq W \geq 1$ we have $q \geq 4$, so G is not a single edge, and the neighbour of a terminal is a non-terminal. Let T be a spanning tree rooted at a non-terminal r; every vertex has at most three children and every terminal is a leaf of T (it has degree one). Write $\rho(Y) = \rho|X \cap Y|$. Maintain a subtree T' containing r, initially T. While $\rho(V(T')) \geq W$, pick a vertex x of T' whose subtree T'_x in T' satisfies $\rho(V(T'_x)) \geq W$ while $\rho(V(T'_y)) < W$ for every child y of x (walk down from r). Since $\rho < 1 \leq W$, x is not a leaf, so it is a non-terminal and $\rho(V(T'_x)) < 3W$. Output $V(T'_x)$ as a cluster and delete T'_x from T' (if $x = r$ the tree becomes empty). When the loop stops, the remaining tree has weight less than W; if it is non-empty, add its vertices to the last output cluster, whose top vertex has its parent in the remaining tree. At least one cluster is output since $\rho q \geq W$, every cluster induces a connected subgraph (it contains a spanning subtree), and the weights are in $[W, 3W)$, or $[W, 4W)$ for the merged cluster. □

From now on fix $W := 2$ and assume $\rho q \geq 2$. Fix the partition of Lemma 3.9, and write $G_j := G[V_j]$, $S_j := X \cap V_j$, and $j(u)$ for the index of the cluster containing u. Thus

$$2/\rho \leq |S_j| < 8/\rho \qquad \text{for every } j. \tag{3}$$

Distinct cluster graphs are vertex-disjoint, hence edge-disjoint. An edge of G whose endpoints lie in different clusters belongs to no cluster graph.

Definition 3.10. Let $s \in S_j$. A *bad bridge for s* is an edge e of G_j that is a bridge of G_j but not a bridge of G, such that the component $Z_e(s)$ of $G_j - e$ containing s satisfies $\rho|S_j \cap Z_e(s)| < 1$. The terminal s is *good* if it has no bad bridge.

Lemma 3.11. *Let $s \in S_j$.*

(i) If s is good, then s spreads in (G_j, S_j).

(ii) If s is not good, there is a bad bridge e_s for s such that $Z_s := Z_{e_s}(s) \subseteq Z_e(s)$ for every bad bridge e for s. Moreover there is an edge $f_s = x_s y_s$ of G with $x_s \in Z_s$ and $y_s \notin V_j$.

Both can be computed in polynomial time.

Proof. (i) follows from Lemma 3.8 with $Q = G_j$ and $R = S_j$, since $\rho|S_j| \geq 2$.

(ii) Let $e \neq e'$ be bad bridges for s, with $Z = Z_e(s)$, $Z' = Z_{e'}(s)$ and far sides $F = V_j \setminus Z$, $F' = V_j \setminus Z'$. We claim $Z \subseteq Z'$ or $Z' \subseteq Z$. The graph $G_j[F]$ is a component of $G_j - e$, so it is

connected. If e' is not an edge of $G_j[F]$, then F lies in one component of $G_j - e'$, so $F \subseteq F'$ (i.e. $Z' \subseteq Z$) or $F \cap F' = \emptyset$. If e' is an edge of $G_j[F]$, then $G_j[Z]$ is connected, contains s and avoids e', so $Z \subseteq Z'$. It remains to exclude $F \cap F' = \emptyset$: then $F' \subseteq Z$, and since $\rho|S_j \cap Z'| < 1$ and $\rho|S_j| \geq 2$ we get $\rho|S_j \cap Z| \geq \rho|S_j \cap F'| > 1$, contradicting that e is bad. Hence the sets $Z_e(s)$ over bad bridges e form a chain; distinct bridges have distinct sides (as $\delta_{G_j}(Z_e(s)) = \{e\}$), and we let e_s be the one with the smallest side.

Since e_s is not a bridge of G and $V_j \setminus Z_s \neq \emptyset$, the cut $\delta_G(Z_s)$ contains an edge $f_s \neq e_s$; write $f_s = x_s y_s$ with $x_s \in Z_s$. If $y_s \in V_j$, then f_s would be an edge of G_j joining the two components of $G_j - e_s$, so $f_s = e_s$; hence $y_s \notin V_j$. □

Lemma 3.12 (tagging)**.** *Let $s \in S_j$ be not good, and let A be a connected subgraph of G that is edge-disjoint from G_j, contains x_s, and contains $S_{j'}$ for some $j' \neq j$. Then s spreads in $(G_j \cup A,\ S_j \cup S_{j'})$.*

Proof. Let $Q = G_j \cup A$ (connected) and $R = S_j \cup S_{j'}$, so $\rho|R| \geq 2$. We verify the hypothesis of Lemma 3.8. Let e be a bridge of Q that is not a bridge of G, let Z be the component of $Q - e$ containing s, and let $Z_0 = Z \cap V_j$. Every edge of G_j between Z_0 and $V_j \setminus Z_0$ lies in $\delta_Q(Z) = \{e\}$. If there is no such edge, connectivity of G_j gives $V_j \subseteq Z$ and $\rho|R \cap Z| \geq \rho|S_j| \geq 2$. Otherwise $e \in E(G_j)$ is a bridge of G_j and $Z_0 = Z_e(s)$. If $\rho|S_j \cap Z_0| \geq 1$ we are done. If not, e is a bad bridge for s, so $x_s \in Z_s \subseteq Z_0 \subseteq Z$ by Lemma 3.11(ii). Since A is connected, contains x_s and does not contain e, we get $V(A) \subseteq Z$, so $\rho|R \cap Z| \geq \rho|S_{j'}| \geq 2$. □

Lemma 3.13 (pairing in trees; cf. [7, Lemma 3.14])**.** *Let H be a tree and Y a multiset of vertices of H of even size. Then one can compute $|Y|/2$ pairwise edge-disjoint paths in H (a path may consist of a single vertex) whose endpoints are exactly the elements of Y, each element used once.*

Proof. Induction on $|V(H)|$. Two copies of the same vertex can be paired by a trivial path, which uses no edge; so we may assume that every vertex occurs at most once in Y. If $|V(H)| = 1$ then $Y = \emptyset$. Otherwise let ℓ be a leaf of H with neighbour u. If $\ell \notin Y$, apply induction to $H - \ell$. If $\ell \in Y$, let Y' be the multiset obtained from Y by replacing ℓ by an (additional) copy of u, apply induction to Y' in $H - \ell$, and extend the path that ends at this copy of u by the edge $u\ell$ (a trivial path at u becomes the path $u\ell$). The edge $u\ell$ is used by exactly one path. □

3.5 Selecting the pairs: proof of Theorem 3.1

If $\rho q < 2$, i.e. $\alpha q < 8$, we route a single pair along a path; this suffices since $|M| \leq q/2$ gives $\alpha|M|/2^{15} < 1$. Assume $\rho q \geq 2$ and fix the clusters as above.

Step 1: one pair per cluster. Scan the pairs of M in any order and add st to a set Z if neither $j(s)$ nor $j(t)$ is marked; then mark $j(s)$ and $j(t)$.

Claim 1. *$|Z| \geq \rho|M|/16$, and every cluster contains terminals of at most one pair of Z.*

Proof. A pair of $M \setminus Z$ has an endpoint in a cluster marked by an earlier pair of Z. Every pair of Z marks two clusters with fewer than $16/\rho$ terminals in total by (3), so fewer than $16/\rho$ pairs of M (including itself) have an endpoint there. Hence $|M| < 16|Z|/\rho$. If a cluster contained terminals of two pairs of Z, the later one would have been added while the cluster was marked. □

Let $Z_1 \subseteq Z$ be the pairs with $j(s) = j(t)$. If $|Z_1| \geq |Z|/2$, route every $st \in Z_1$ along a path in $G_{j(s)}$; these paths are edge-disjoint by Claim 1, and $|Z_1| \geq \rho|M|/32$. Otherwise let $Z' = Z \setminus Z_1$, so $|Z'| \geq |Z|/2$. For a terminal s of a pair of Z' we call $j(s)$ its *own cluster*; by Claim 1, distinct terminals of pairs of Z' have distinct own clusters, and an own cluster contains no other terminal of a pair of Z.

Step 2: tags and an independent set. For every terminal s of a pair of Z' that is not good (in its own cluster), Lemma 3.11 gives $f_s = x_s y_s$; let $\tau(s) := j(y_s) \neq j(s)$ be the cluster *tagged* by s. Let D be the digraph on Z' with an arc from P to $P' \neq P$ whenever a non-good terminal s of P has $\tau(s) = j(u)$ for a terminal u of P'. Every pair has out-degree at most two, because a cluster contains terminals of at most one pair of Z.

Claim 2. *One can compute $I \subseteq Z'$ with $|I| \geq |Z'|/5$ such that no arc of D joins two pairs of I.*

Proof. Every subset $Z'' \subseteq Z'$ induces at most $2|Z''|$ arcs, so it contains a pair with at most four neighbours (in- or out-) within Z''. Repeatedly put such a pair into I and delete it and its neighbours. □

Let $I_2 \subseteq I$ be the pairs st with $\tau(s) = j(t)$ or $\tau(t) = j(s)$, and $I_3 = I \setminus I_2$. Suppose $|I_2| \geq |I|/2$. For $st \in I_2$ with, say, $\tau(s) = j(t)$, take a walk consisting of a path from s to x_s in $G_{j(s)}$, the edge f_s, and a path from y_s to t in $G_{j(t)}$. All edges of this walk have both endpoints in $V_{j(s)} \cup V_{j(t)}$, and for distinct pairs of I_2 these unions are disjoint. So the walks are edge-disjoint, I_2 is routable, and $|I_2| \geq |I|/2 \geq \rho|M|/320$. From now on assume $|I_3| \geq |I|/2$.

Claim 3. *If s is a non-good terminal of a pair of I_3, then the cluster $\tau(s)$ contains no terminal of any pair of I.*

Proof. Let $st \in I_3$. We have $\tau(s) \neq j(s)$, and $\tau(s) \neq j(t)$ since $st \notin I_2$. If $\tau(s) = j(u)$ for a terminal u of a pair $P' \in I \setminus \{st\}$, then D has an arc from st to P', contradicting Claim 2. □

The clusters $\tau(s)$ of Claim 3 are called *transit clusters*. They contain no terminal of I, so a transit cluster is nobody's own cluster. For a transit cluster C let $T(C)$ be the set of non-good terminals s of pairs of I_3 with $\tau(s) = C$, and $d(C) = |T(C)|$.

The tag edges are pairwise distinct and belong to no cluster graph: f_s joins the own cluster $j(s)$ to a transit cluster, so its endpoint in a non-transit cluster is x_s; if $f_s = f_u$ for terminals $s \neq u$ of pairs of I_3, then $x_s = x_u$, so $j(s) = j(u)$, which is impossible for distinct terminals.

Step 3: pairing through transit clusters. For every transit cluster C with $d(C) \geq 2$, fix a spanning tree H_C of G_C; if $d(C)$ is odd, declare one terminal of $T(C)$ *isolated*. Apply Lemma 3.13 to the multiset $\{y_s : s \in T(C) \text{ not isolated}\}$ in H_C; if a path P pairs y_s with $y_{s'}$, call s and s' *partners* and set $P_{ss'} := P$. Let B be the graph on the terminals whose edges join partners; every terminal has degree at most one in B.

Claim 4. *The set Y of pairs of I_3 with no isolated terminal satisfies $|Y| \geq |I_3|/3$.*

Proof. A transit cluster has at most one isolated terminal, and only if $d(C) \geq 3$. Since every terminal lies in at most one set $T(C)$, the number of isolated terminals is at most $\frac{1}{3}\sum_C d(C) \leq \frac{2}{3}|I_3|$, and each of them excludes one pair. □

Step 4: sparsification. Two distinct pairs of Y *conflict* if a terminal of one is a partner of a terminal of the other. Each pair conflicts with at most two others, so greedily we find a conflict-free $Y' \subseteq Y$ with $|Y'| \geq |Y|/3$. Let Y'_a be the pairs $st \in Y'$ in which s and t are partners, and $Y'_b = Y' \setminus Y'_a$.

Suppose $|Y'_a| \geq |Y'|/2$. For $st \in Y'_a$ partnered through C, take the walk formed by a path from s to x_s in $G_{j(s)}$, the edge f_s, the path P_{st} in H_C, the edge f_t, and a path from x_t to t in $G_{j(t)}$. Own clusters are distinct and are not transit clusters, pairing paths in one tree H_C are edge-disjoint, distinct transit clusters are vertex-disjoint, and tag edges are distinct and lie in no cluster graph. Hence these walks are edge-disjoint and Y'_a is routable.

Step 5: spreading for Y'_b. Now suppose $|Y'_b| \geq |Y'|/2$.

Claim 5. *Let s be a terminal of a pair $st \in Y'_b$. Then exactly one of the following holds: (α) s is good; (β) s is not good and $d(\tau(s)) = 1$; (γ) s is not good, $d(\tau(s)) \geq 2$, and s has a partner s' that is not a terminal of any pair of Y'.*

Proof. If s is not good and $d(\tau(s)) \geq 2$, then s is not isolated (as $st \in Y$), so it has a partner s'. We have $s' \neq t$ because $st \notin Y'_a$, and if s' were a terminal of a pair $P' \in Y'$, then $P' \neq st$ and st would conflict with P'. □

For every terminal s of a pair of Y'_b define a subgraph $Q(s)$ and a receiver set $R(s)$:
(α) $Q(s) = G_{j(s)}$ and $R(s) = S_{j(s)}$;
(β) with $C = \tau(s)$: $Q(s) = G_{j(s)} \cup \{f_s\} \cup G_C$ and $R(s) = S_{j(s)} \cup S_C$;
(γ) with $C = \tau(s)$: $Q(s) = G_{j(s)} \cup \{f_s\} \cup P_{ss'} \cup \{f_{s'}\} \cup G_{j(s')}$ and $R(s) = S_{j(s)} \cup S_{j(s')}$. Here $j(s')$ is the own cluster of s'; we call it the *donor cluster* of s.

Claim 6. *Every such s spreads in $(Q(s), R(s))$.*

Proof. Case (α) is Lemma 3.11(i). In case (β) apply Lemma 3.12 with $A = \{f_s\} \cup G_C$: it is connected (f_s joins x_s to $y_s \in V_C$), contains x_s and S_C, and is edge-disjoint from $G_{j(s)}$ because $V_C \cap V_{j(s)} = \emptyset$. In case ($\gamma$) apply Lemma 3.12 with $A = \{f_s\} \cup P_{ss'} \cup \{f_{s'}\} \cup G_{j(s')}$: it is connected (the walk $x_s, y_s, P_{ss'}, y_{s'}, x_{s'}$ ends in $V_{j(s')}$), contains x_s and $S_{j(s')}$, and $j(s') \neq j(s)$ since $s' \notin \{s, t\}$. It is edge-disjoint from $G_{j(s)}$: f_s and $f_{s'}$ have an endpoint in V_C, $P_{ss'} \subseteq G_C$, and $V_{j(s')} \cap V_{j(s)} = \emptyset$. □

Claim 7. *For distinct terminals $s \neq u$ of pairs of Y'_b, the graphs $Q(s)$ and $Q(u)$ are edge-disjoint and $R(s) \cap R(u) = \emptyset$.*

Proof. Each $Q(s)$ consists of: its own cluster graph; at most two tag edges; in case (β) a transit cluster graph G_C with $d(C) = 1$; in case (γ) a pairing path inside a transit cluster with $d(C) \geq 2$ and a donor cluster graph. We check that no part is shared. Own clusters of distinct terminals are distinct. A donor cluster $j(s')$ contains the terminal s' of a pair of I that is not in Y'; it is therefore not the own cluster of any terminal of Y'_b (an own cluster contains terminals of only one pair of Z) and not a transit cluster. The donors of distinct terminals are distinct (partners are unique), and the donor clusters of distinct donors s', u' are distinct: otherwise s', u' would be the two terminals of one pair of Z', which lie in different clusters. A transit cluster C with $d(C) = 1$ is used, in case (β), only by the unique terminal of $T(C)$, and it carries no pairing path. Pairing paths in one tree are edge-disjoint, and $P_{ss'}$ is used only by s, since its other end s' is not a terminal of Y'. Tag edges are distinct, lie in no cluster graph, and $f_{s'}$ is used only by the unique partner s of s'. Finally, $R(s)$ is the union of S_j over the own cluster and the transit or donor cluster of s; we showed that these clusters are distinct for distinct terminals, and the sets S_j are disjoint. □

For every terminal s of a pair of Y'_b let g_s be the flow given by Claim 6. By Claim 7, the total load of these flows is at most $1/2$ on every non-bridge and at most one on every bridge, and every $v \in X$ receives at most ρ, from at most one terminal. Thus they form a spreading system, and Lemma 3.6 routes Y'_b with congestion one.

Counting. In every branch the routed set has size at least $|Y'|/2 \geq |Y|/6 \geq |I_3|/18 \geq |I|/36 \geq |Z'|/180 \geq |Z|/360 \geq \rho|M|/5760 = \alpha|M|/23040$, and $23040 < 2^{15}$; the other branches give more. If the routed set happens to be empty (possible only when $\alpha|M|/2^{15} < 1$), we route a single pair instead. Every step (clusters, bridges, tags, greedy selections, pairing, maximum flows, the final linear program and cycle cancelling) takes polynomial time. By Lemma 3.3(d) the routing transfers to the original graph. This proves Theorem 3.1. □

3.6 Non-uniform paired weights: proof of Corollary 3.2

If $\pi(X) < 2^{19}$, some pair has positive weight, so its endpoints are connected (the product demand between them is positive); route that pair alone. Assume $\pi(X) \geq 2^{19}$ and let $q = |X|$ and $\phi_0 = \pi(X)/(4q)$. Discard all pairs with $\pi(s) < \phi_0$; their total weight is less than $q\phi_0 = \pi(X)/4$. For each remaining terminal u let $n(u) = \lfloor \pi(u)/\phi_0 \rfloor \geq 1$ and $\pi'(u) = \phi_0 n(u) \geq \pi(u)/2$, and $\pi' = 0$ on discarded terminals. Then $\pi'(X) \geq \frac{3}{8}\pi(X)$ and $n(u) \leq 1/\phi_0 \leq 4q$. Let $\lambda = \pi'(X)/\pi(X) \in [\frac{3}{8}, 1]$ and $\pi'' = \lambda\pi'$. Since $\pi''(u)\pi''(v)/\pi''(X) = \pi'(u)\pi'(v)/\pi(X) \leq \pi(u)\pi(v)/\pi(X)$, the set $X'' = \mathrm{supp}(\pi'')$ is π''-flow-linked, and $\pi''(X) \geq \frac{9}{64}\pi(X)$.

Construct G' from G by attaching, to every $u \in X''$, $n(u)$ new pendant vertices (*copies* of u), as in [6, Lemma 1.1]. For each remaining pair st (note $n(s) = n(t)$) match the r-th copy of s with the r-th copy of t; let M_c be the resulting $q' = \sum_u n(u)$ copies' matching. Put $\phi = \lambda\phi_0 \leq \frac{1}{4}$, so that $\pi''(u) = \phi\, n(u)$ and $\pi''(X) = \phi q'$. We claim the set of copies is ϕ-flow-linked in G'. Between copies of distinct u, v the total demand is $n(u)n(v)\phi/q' = \pi''(u)\pi''(v)/\pi''(X)$, which we route along the π''-product flow between u and v (at most the π-product flow) extended by the pendant edges; between two copies of the same u we route through u. Each pendant edge carries at most $\phi(q'-1)/q' < 1$, and the original edges carry at most one. All copies lie in one component of G', to which we apply Theorem 3.1 with $\alpha = \phi$. It returns at least $\phi|M_c|/2^{15} = \pi''(X)/2^{16} \geq 9\pi(X)/2^{22} \geq \pi(X)/2^{19}$ copy pairs routable with congestion one.

No two of them are copies of the same pair st: the unique edge of G at s separates s and all its copies from t and all its copies, so two such copy pairs would each send one unit across this edge. Deleting the pendant edges from the routing (after cycle cancelling, a path from a copy of s to a copy of t is a path from s to t extended by two pendant edges) gives a routing of the corresponding pairs of M in G with congestion one. All quantities are polynomial since $n(u) \leq 4q$. □

4 Well-linked decomposition with near-logarithmic loss

Theorem 4.1. *Let G be a graph and $(s_i, t_i)_{i\in[k]}$ pairs whose $2k$ endpoints are distinct vertices of degree one. Let f be a feasible multicommodity flow routing $x_i \in [0,1]$ between s_i and t_i, with $F = \sum_i x_i > 0$. Then one can compute in polynomial time, using a constant number of calls to the algorithm of Theorem 2.3 on G with an integral weighting of total weight at most $4k$, disjoint sets $U_1, \ldots, U_\ell \subseteq V(G)$, and for every j a set P_j of pairs with both endpoints in U_j and weights π_j on the endpoint set X_j of P_j, such that:*

- *(i) $\pi_j(s_i) = \pi_j(t_i) \leq 1/8$ for $i \in P_j$; moreover all values of all π_j are positive integer multiples of a common rational $\phi > 0$;*
- *(ii) every vertex of X_j has degree one in $G[U_j]$, and X_j is π_j-flow-linked in $G[U_j]$;*
- *(iii) $\sum_j \pi_j(X_j) \geq 3F/(128\,\Gamma(4k))$.*

Proof. Discretization. Let $\varepsilon = F/(2k)$. For every i let $d_i = \lfloor x_i/\varepsilon \rfloor$ if $x_i \geq \varepsilon$ and $d_i = 0$ otherwise (this is the uniformization idea of [6, Lemma 1.1], at granularity ε). Let $K = \sum_i d_i$. Then $K \leq \sum_i x_i/\varepsilon = 2k$. Moreover $\lfloor y \rfloor \geq y/2$ for $y \geq 1$, and the pairs with $x_i < \varepsilon$ have total value less than $k\varepsilon = F/2$, so

$$K\varepsilon \;\geq\; \tfrac{1}{2}\textstyle\sum_{i:x_i\geq\varepsilon} x_i \;\geq\; F/4. \tag{4}$$

Let $A(s_i) = A(t_i) = d_i$ for every i and $A = 0$ elsewhere; this is well defined because the endpoints are distinct, and $|A| = 2K \leq 4k$.

One decomposition. With $\Gamma = c_0\Gamma_1$ as in Section 2, for $j = 0, 1, 2, \ldots$ run the algorithm of Theorem 2.3 on (G, A) with $\phi_j = \varepsilon/(2^{j+3}\Gamma_1(2K))$, obtaining C_j, and stop at the first j with

$|C_j| \le \varepsilon K/4$. Since $|C_j| \le \phi_j \cdot 2K \cdot c_0\Gamma_1(2K) = c_0\varepsilon K/2^{j+2}$, the search stops at some $j \le \lceil \log c_0 \rceil$, i.e. after $O(1)$ calls. Let ϕ and C be the final values. Then

$$|C| \le \varepsilon K/4, \qquad \phi \;\ge\; \frac{\varepsilon}{2^{\lceil \log c_0 \rceil + 3}\Gamma_1(2K)} \;\ge\; \frac{\varepsilon}{16\,\Gamma(2K)},$$

and $A|_U$ is ϕ-flow-expanding in $G[U]$ for every component U of $G - C$. (If c_0 is known, a single call with $\phi = \varepsilon/(8\Gamma(2K))$ suffices.)

Charging. Call pair i (with $d_i \ge 1$) *separated* if s_i and t_i lie in different components of $G - C$. If Z is the component containing s_i, then $\delta_G(Z) \subseteq C$, so the flow of pair i in f has load at least $x_i \ge \varepsilon d_i$ on C. As f has load at most one on each edge (cf. [6, Lemma 2.2]),

$$\varepsilon \textstyle\sum_{i \text{ separated}} d_i \;\le\; \sum_{i \text{ separated}} x_i \;\le\; |C| \;\le\; \varepsilon K/4.$$

Hence the pairs that are not separated and have $d_i \ge 1$ (the *surviving* pairs) satisfy $\sum_{i \text{ surviving}} d_i \ge 3K/4$.

The pieces. For every component U of $G - C$ that contains a surviving pair, let P_U be the set of surviving pairs with both endpoints in U, X_U their endpoints, and $\pi_U = \phi A|_{X_U}$. Property (i) holds since $\phi \le \varepsilon/(8\Gamma_1(2K)) \le \varepsilon/8$ gives $\phi d_i \le \phi x_i/\varepsilon \le 1/8$, and all values are multiples of ϕ. For a surviving pair i in U, the vertices $s_i \ne t_i$ are connected in $G - C$, so the unique edge at s_i is not in C and the neighbour of s_i lies in U; thus s_i (and likewise t_i) has degree one in $G[U]$. By Lemma 2.4, applied to the ϕ-flow-expanding weighting $A|_U$ in $G[U]$ and $Y = X_U$, the set X_U is π_U-flow-linked in $G[U]$. This proves (ii). Distinct components are disjoint. Finally, by (4),

$$\sum_U \pi_U(X_U) = 2\phi \sum_{i \text{ surviving}} d_i \;\ge\; \tfrac{3}{2}\phi K \;\ge\; \frac{3K\varepsilon}{32\,\Gamma(2K)} \;\ge\; \frac{3F}{128\,\Gamma(4k)},$$

using monotonicity of Γ. All steps are polynomial. □

Remark 4.2 (other decomposition algorithms)**.** The proof uses only the following property of Theorem 2.3: for the integral weighting A with $|A| \le 4k$ and any $\phi > 0$, the algorithm returns C with $|C| \le \phi|A|\gamma$ such that $A|_U$ is ϕ-flow-expanding in $G[U]$ for every component U of $G - C$. Any algorithm with this property, for a known γ, yields Theorem 4.1 with $16\Gamma(4k)$ replaced by 8γ (take $\phi = \varepsilon/(8\gamma)$).

5 The approximation algorithm

Proof of Theorem 1.1. Normalize the instance (Section 2) and solve the LP (1) in its compact edge form; let x and f be an optimal solution. If LP $= 0$, no pair is connected and we return the empty set. Otherwise LP $\ge$ OPT ≥ 1, since a single connected pair is routable. Apply Theorem 4.1 to f with $F = \mathrm{LP}$. For every piece (U_j, P_j, π_j) apply Corollary 3.2 in $G[U_j]$; its hypotheses hold by Theorem 4.1(i)–(ii). It returns a set of pairs of P_j of size at least $\pi_j(X_j)/2^{19}$ routable in $G[U_j]$ with congestion one. The sets U_j are disjoint, so the induced subgraphs are edge-disjoint and the union of these routings is a routing of congestion one in G. Its size is at least

$$\sum_j \frac{\pi_j(X_j)}{2^{19}} \ge \frac{3\,\mathrm{LP}}{2^{26}\,\Gamma(4k)}.$$

(If no piece exists, we return one connected pair; Corollary 3.2 returns at least one pair per piece, so the output is never empty.) All steps other than the calls to the algorithm of [4] are deterministic and polynomial. By (2), $\Gamma(4k) \le C' \log(k+2) \exp(C'\sqrt{\log\log(k+4)})$ for an absolute constant C', and LP $\ge$ OPT. □

Corollary 5.1 (integrality gap)**.** *For every unit-capacity instance with k pairs,* $\mathrm{LP} \le O(\Gamma(4k)) \cdot \mathrm{OPT} = \log^{1+o(1)} k \cdot \mathrm{OPT}$.

Remark 5.2 (integer capacities)**.** If edges have integer capacities $u(e)$ and demands are one, one can replace every edge by $u(e)$ parallel unit edges. To keep the instance polynomial, first use the argument of [7, Section 2]: decompose an optimal edge flow into polynomially many paths and take a basic optimal solution of the path formulation restricted to these paths. A pair with $x_i \in (0,1)$ has a non-tight pair constraint, so the number of such pairs is at most the number of tight edge constraints, i.e. at most m. If the pairs with $x_i = 1$ carry at least half of LP, they form a routable set; otherwise restrict to the at most m fractional pairs, whose total load on any edge is at most m, and cap the capacities at m. This loses a factor two, and Theorem 1.1 then applies with the same asymptotic ratio.

Remark 5.3 (weighted version, with an extra logarithm)**.** For pairs with profits $w_i \ge 0$ and the weighted relaxation $\mathrm{LP}_w = \max \sum_i w_i x_i$, the standard class argument gives ratio $O(\Gamma(4k) \log k) = \log^{2+o(1)} k$. Let W be the largest profit of a pair with connected endpoints, so $\mathrm{LP}_w \ge W$ (pairs with disconnected endpoints have $x_i = 0$). Pairs with $w_i < W/k^2$ contribute at most $W/k \le \mathrm{LP}_w/k$ and are discarded; the others fall into $O(\log k)$ classes $\{i : w_i \in (W2^{-j-1}, W2^{-j}]\}$, one of which carries at least an $\Omega(1/\log k)$ fraction of LP_w. Theorem 4.1 and Corollary 3.2 apply to the optimal flow restricted to that class (any feasible flow may be used), and every routed pair has profit at least half the maximum profit in its class. We do not know how to avoid the extra logarithm; see Section 6.

Remark 5.4 (Anand's decomposition; conditional)**.** Anand [1] improves flow-expander decomposition to overhead $O(\log n \log\log n)$. His node-weighted theorem (Theorem 5.1 of [1]) gives overhead $O(\log |A| \log\log |A|)$ for rational weightings with $A(v) \ge 1$ for *every* vertex v, so that $|A| \ge n$. Our weighting vanishes outside the $2k$ terminals, so this bound does not give an overhead in terms of k. For weightings that may vanish, Section 5 and Appendix C of [1] state overhead $O(\log n \log\log n)$, with n the number of vertices; we have not verified that appendix. *If* that statement holds, Remark 4.2 and Corollary 3.2 give a polynomial-time $O(\log n \log\log n)$-approximation for AN-MCF (the normalized graph has $n+2k$ vertices, and by the previous remark we may assume $k \le m$). This bound is incomparable with Theorem 1.1 and better when $n \le k^{O(1)}$. Obtaining $O(\log k \log\log k)$ would require a node-weighted decomposition whose overhead depends only on the total weight even when most vertices have weight zero, or a reduction eliminating zero-weight vertices without loss; we leave this open.

6 Lower bounds and open problems

Lower bounds. Andrews, Chuzhoy, Guruswami, Khanna, Talwar and Zhang [2] show that AN-MCF is hard to approximate within $\log^{1/2-\delta} n$ unless NP $\subseteq$ ZPTIME($n^{\mathrm{polylog}\, n}$), and that the integrality gap of (1) is $\log^{1/2-\delta} n$ (as quoted in [7]). These bounds are stated in terms of the number n of vertices. Pairs with $s_i = t_i$ count equally in LP and OPT and can be removed; afterwards, keeping at most $\deg(u)$ copies of each repeated pair $\{u, v\}$ changes neither LP nor OPT, so one may assume $k \le n^2 m$. For instances of polynomial size ($m \le n^{O(1)}$, as produced by such reductions) this gives $\log k = O(\log n)$, and the lower bounds read $\Omega(\log^{1/2-\delta} k)$. Thus on such instances the integrality gap of the natural relaxation lies between $\log^{1/2-\delta} k$ and $\log^{1+o(1)} k$. The $\Omega(\log n)$ lower bound on the overhead of expander decompositions [4] concerns the decomposition, not AN-MCF, and we do not know whether $\Omega(\log k)$ is a barrier for approaches based on flow-linked decompositions.

Weighted version. For pairs with profits w_i, the decomposition step charges separated pairs by the amount of flow crossing the cut, which bounds their total *flow*; a cut of the allowed size may separate all of the most profitable pairs. Weighting the terminals by profit makes the cut bound grow with the profit range, and grouping pairs into profit classes loses a further logarithm, which gives the $\log^{2+o(1)} k$ bound of Remark 5.3. We do not know how to obtain a $\log^{1+o(1)} k$ ratio for weighted AN-MCF. (We expect the rounding of Section 3 to extend to weights along the lines indicated in [7]; the obstacle is the decomposition.)

Node capacities. For node-capacitated AN-MCF, [6] obtain polylogarithmic ratios with congestion $1 + \varepsilon$. An analogue of Theorem 1.1 would require vertex flow-expander decompositions with near-logarithmic overhead for integral weightings, which [4] list as open, and a node analogue of the rounding.

Other questions. Can the ratio be improved to $O(\log k)$, or even to $O(\log^{1/2+o(1)} k)$? Can the decomposition of Theorem 4.1 improve other applications of well-linked decompositions in general graphs, such as those in [9]?

Computational sanity checks. We implemented the rounding of Section 3 (degree reduction, clusters, bad bridges, tags, Steps 1–5, spreading flows by maximum flow) and ran it on 270 random instances with up to 68 terminals, checking every intermediate assertion of the proof (nesting of bad bridges, existence of tag edges, transit clusters free of terminals of I, edge-disjointness of the graphs $Q(s)$ and of the direct routings, the spreading-system conditions) and verifying by linear programming that each selected set is routable with congestion one. Lemmas 3.8, 3.11 and 3.12 were also tested directly on 20000 random small graphs. Because of the constants, small instances mostly exercise the simpler branches of Section 3.5: the branches Z_1, I_2 and Y_b' (with cases (α) and (β)) occurred, while case (γ) and the branch Y_a' did not occur (Lemma 3.12, which case (γ) uses, was tested directly). These checks are diagnostics only and are not part of any proof.

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